

To get the top right corner of x-center,

First calculate the x-center coordinate with respect to the dimension provided.

You will get x-center coordinate. Then, Normalized width, take it and multiply with the dimension.

Now you will be having both x-center coordinate and x-center width.

$$X_{TOPLEFT} = x_{CENTER} - \frac{WIDTH}{2}$$

$$Y_{TOPLEFT} = Y_{CENTER} - \frac{HEIGHT}{2}$$

$$X_{TOPRIGHT} = x_{CENTER} + \frac{WIDTH}{2}$$

$$Y_{TOPLEFT} = Y_{CENTER} - \frac{HEIGHT}{2}$$

When dilation is added into convolution, then the output matrix size calculation becomes,

$$Output_{dimension} = \frac{W - Dilation(K-1) - 1 + 2P}{S} + 1$$

Effective receptive field size would be,

$$Dilation \times (K-1) + 1$$

Slow RCNN:

It uses external RoI, then does multiple cnn. First does ROI and then creates the feature map.

Fast RCNN

First feature map is created. No anchor boxes. Uses external region proposal net.

Faster RCNN

Anchor boxes are there. But has inbuilt RPN.

Yolo:

No anchor boxes. Uses Bounding box. Per cell its 5*B (classid, x, y, w, h). The model uses separate components within its detection head for classification and bounding box regression.

Consider the task of applying a convolutional layer with a **stride of 1** to an image of size 111×111 (Grayscale), while ensuring that the **output feature map retains the same spatial dimensions**. Which of the following code snippets correctly implements this using PyTorch?

OPTIONS :

- ☐ `nn.Conv2d(in_channels=1, out_channels=64, kernel_size=3, stride=1, padding=1)` 
- ☐ `nn.Conv2d(in_channels=1, out_channels=64, kernel_size=3, stride=1, padding=2)` 
- ☐ `nn.Conv2d(in_channels=1, out_channels=64, kernel_size=5, stride=1, padding=1)` 
- ☒ `nn.Conv2d(in_channels=1, out_channels=64, kernel_size=1, stride=1, padding=0)` 

For this question, first option is marked as right. But I get last option also right. Here is my calculation.

The formula is, $\frac{W - K + 2P}{S} + 1$

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for option 2, $\frac{111 - 1 + (2 \times 0)}{1} + 1$

for option 1, $\frac{111 - 3 + (2 \times 1)}{1} + 1$

$$110 + 1 = 111$$

$$113 - 3 + 1 = 111$$